Mission 2: FST Dialogue Systems, Lovelace's Objection and Noisy Channels

Solution documentation, Squadron Bengio

- Course: Conversational AI: History, Applications, Future (BTU Cottbus-Senftenberg, WS 2026/27)
- Problem set: ps02, Week 2
- Squadron: Bengio (Moodle Group 11)
- Repository: https://github.com/fahadakram009/squadron-bengio-cai (folder `mission-02`)
- Date: 20 October 2026

1. Mission objective

As the USSE Noosphere passes the Sirius binary system, plasma noise corrupts AURA's subspace transceiver, and Command suspects AURA may only be manipulating symbols without understanding. The mission applies three foundations to test this: deterministic Finite State Transducers (FST) for dialogue control, Lovelace's objection and Turing's imitation game for machine originality, and Shannon's information theory for communication under noise. The deliverable is a runnable Python simulator that joins an FSTDialogueManager with five states and slot memory to a noisy channel allowing at most seven words per turn with a mandatory noise token, a five-question Anti-Turing interrogation of a modern LLM, and a ten-turn noisy interrogation transcript, backed by three design documents, one per officer station.

2. Crew and stations

Station	Officer	Deliverable	Status
Historical Archivist (Data and Payload Specialist)	Osaf Ahmad	01_Lovelace_Inquiry_Brief.md and .pdf	signed off
Protocol Specialist (Communications and Protocol Officer)	Madhurima Lakshmi Prabha	02_FST_and_Noisy_Protocol.md	signed off
Ethical Navigator (Security and Ethics Sentinel)	Janusz Manczak	03_Entropy_Hazard_Audit.md	signed off
Core Architect (Systems Architect)	Fahad Akram	04_fst_noisy_simulator_starter.py, 05_Noisy_Interrogation_Transcript.md, repository, logs	signed off

All four officers delivered within two days of the mission start. The stations interlock more tightly than in Mission 1: the Protocol Specialist audited the Architect's code and the Architect applied her fixes, the Ethical Navigator ran his stress tests through the simulator and quoted its output, and the simulator reads the Archivist's questions and the Navigator's test utterances at run time.

3. Repository layout and how to run

File	Content
README.md	mission overview, run instructions, station integration table, sign-off checklist
ps02_briefing.md	the instructor's mission specification
01_Lovelace_Inquiry_Brief.md, .pdf	Station 01 deliverable
02_FST_and_Noisy_Protocol.md	Station 02 deliverable
03_Entropy_Hazard_Audit.md	Station 03 deliverable
04_fst_noisy_simulator_starter.py	Station 04 deliverable: the simulator, single file
05_Noisy_Interrogation_Transcript.md	squad deliverable (Part 5), generated by the simulator from a live run
logs/20261019-234831-ps02-all.*	committed transcript of that live run (JSONL log, result file, console output)
requirements.txt	Python dependencies

Running the simulator:

```
source .venv/bin/activate && pip install -r mission-02/requirements.txt
cd mission-02
python 04_fst_noisy_simulator_starter.py --selftest        # offline, 21 checks
python 04_fst_noisy_simulator_starter.py --mode stress     # FST brittleness, offline
python 04_fst_noisy_simulator_starter.py                   # live run, all modes
python 04_fst_noisy_simulator_starter.py --check-signoffs  # sign-off gate
```

Options: `--mode fst|stress|channel|interrogation|all`, `--turns`, `--seed`, `--strict`, `--provider`, `--model`. API keys are read from a git-ignored `.env` file and are never committed.

4. Station 01: The Lovelace Inquiry Brief (Osaf Ahmad)

The brief frames Lovelace's objection as the claim that a machine follows the operations it was instructed to perform and does not originate ideas, and designs five questions whose answers can be observed without claiming to prove consciousness. Each question targets one capability: constrained invention (a communication ritual with only three symbols), transfer between domains (error-correcting codes applied to misunderstandings between friends), self-critique and revision, counterfactual reasoning (a model that can only follow rules written at deployment), and originality with provenance (a novel combination plus the evidence needed to show it was not memorised).

The protocol was executed against ChatGPT and each response is recorded with an evaluation. A rubric scores constraint adherence, coherence, critical reflection and originality evidence from 0 to 2; every answer scored 7 of 8, losing the point for originality because novelty was plausible but not independently verified. The conclusion is careful: the results challenge a simplistic view of machines as repeating fixed responses but do not settle Lovelace's deeper objection, since a fluent answer does not show understanding or verified novelty. A limitations section and a replication template close the brief. The document was submitted as a PDF; a Markdown version with identical text was added for the grader.

5. Station 02: FST and Noisy Channel Protocol (Madhurima Lakshmi Prabha)

The protocol map gives the FSTDialogueManager as a formal Mealy transducer with the five states INIT, GREETING, COLLECT_SLOTS, CONFIRMATION and TERMINAL plus a recoverable FALLBACK, a state diagram, a transition table, the slot memory buffer with the three slots sector, warp factor and clearance code, pseudocode and a worked example. The noisy channel section fixes four constraints: at most seven words per turn, exactly one noise token from the seven-word vocabulary, a deterministic noise model with erasure rate one in seven, and a validator. It reasons about where the noise should go in terms of mutual information: overwriting a filler word loses almost nothing, overwriting a slot value loses all of that slot. Four signal-reduction rules for free-text turns follow: strip padding, keep the claim and drop the reasons, give the noise token a free slot, and add cheap redundancy, with a worked example that compresses Lovelace's objection from twenty words to seven.

Section 7 is a compliance check of the squad's code. It confirms the state machine is sound and reports five protocol findings: the FST's own replies were up to eighteen words long and never passed through the channel; random substitution could erase slot values; the greeting check matched inside other words such as "this"; sector and warp parsing accepted false positives such as "console" and "I have 2 questions"; and confirmation stored the clearance code on a bare "yes". Each finding comes with a fix, including seven-word reply templates and a noise-placement method that overwrites the first filler word.

6. Station 03: Entropy and Hazard Audit (Janusz Manczak)

The audit formalises the FST as a Mealy tuple and states the brittleness problem: an undefined transition means catastrophic failure. It defines five non-canonical utterances, each quoted so the simulator can read them: all slots offered at once before greeting, a sarcastic reply instead of a sector, an off-topic question about the thrusters, a misspelt sector "Alpha Centori", and a sector given with "standard cruise" instead of a warp number. Running them through the simulator gives a break rate of 80 percent, and 100 percent in strict mode, which replicates the portal's zero-recovery workbench.

The information-theoretic section defines Shannon entropy, mutual information and the erasure channel, then audits the live ten-turn interrogation: source entropy 7.54 bits per word against 5.88 transmitted, a capacity bound of 0.857, and the paradox that lexical retention is low while judged fidelity is high. The hazard section identifies three failure classes, the brittle lockout, the semantic hallucination gap and the Lovelace deception, and tabulates them as a failure modes and effects analysis with severity, probability and mitigation. It closes with a neuro-symbolic guardian architecture in which an LLM parses noisy natural language into typed slot candidates and a deterministic FST acts as the safety kernel that never processes raw strings.

7. Station 04: the FST and Noisy Channel Simulator (Fahad Akram)

The simulator keeps the starter file's public surface, `FSTDialogueManager.process_utterance` and `NoisyChannelSimulator.apply_channel_constraints` with the fixed `NOISE_VOCAB`, and implements the four mission pillars in one file.

7.1 FST dialogue manager

The manager is a deterministic Mealy transducer. Its input alphabet is the set of intent tags a per-state parser produces from the utterance, using whole-word matching against a canonical vocabulary that mirrors the course portal's interactive workbench: greetings, the sectors Orion Arm, Alpha Centauri, Sol System, Sirius Binary and Proxima Centauri, warp factors one to nine, and the clearance code ALPHA-7. The transition function moves INIT to GREETING on a greeting, GREETING to COLLECT_SLOTS on a sector, COLLECT_SLOTS to CONFIRMATION on a warp factor, and CONFIRMATION to TERMINAL only when the clearance code itself is spoken; a bare "yes" asks for the code and "no" aborts to GREETING with the slots cleared. Any unparseable input enters FALLBACK, which remembers the state it left and recovers transparently on the next parseable input; three consecutive failures halt the machine, and the `--strict` flag reproduces the portal's catastrophic failure on the first failure instead. Every verbalisation is at most seven words with one filler word, so the FST's own turns satisfy the channel constraint, and the state-sequence entropy of a canonical run is exactly two bits. A full transition trace with slot updates and fallback counters is recorded per turn.

7.2 Noisy channel

The channel implements Shannon's pipeline. A valid seven-word transmission with a noise token passes unchanged; anything else is re-encoded with the Protocol Specialist's placement rule: truncate to seven words, overwrite the first filler word with a noise token, otherwise append it, and only overwrite the last word when the turn is full. Per-turn metrics report the word counts, the number of noise tokens and the resulting erasure rate, the Shannon entropy of the source and transmitted word streams, the erasure-channel capacity bound $C = 1 - p$, the number of payload slots, and lexical retention, the share of the source's content words that survived.

7.3 Noisy Turing interrogation

An LLM plays AURA. For each of ten questions the model first answers freely as the source, then re-encodes its own answer as a transmitter under the seven-word rule, the channel validates or enforces the constraint, and after all turns a separate receiver decodes every transmission, rates how much of the original meaning survived and how human it reads, and gives a HUMAN or MACHINE verdict. The first five questions are the Archivist's prompts, read from the brief; the remaining five are built in and cover a sycophancy trap, a paradox loop and a hallucination bait. The run writes the squad's Part 5 transcript with a per-turn table, the source answers and decoded meanings, and the channel metrics.

7.4 Stress tests, providers, verification

The stress mode runs the Ethical Navigator's five quoted utterances, or a built-in set when his file is absent, and reports the final state, fallback count, surviving slots and break rate. Provider adapters exist for Anthropic, Google Gemini and OpenAI plus a deterministic mock; the Gemini adapter asks for low thinking so that hidden reasoning does not truncate the output, retries on rate limits and falls back across a pool of models when a free-tier daily quota is exhausted. Each run writes a JSONL log and a result file. The embedded verification function runs 21 offline checks covering the canonical path, fallback recovery and halting, strict mode, abort, each of the five protocol findings, the channel constraint, the entropy function, the metrics, the stress set, question loading and a mock interrogation. The sign-off gate parses the README checklist.

8. Live run results

A full run was executed on 19 and 20 October 2026 with a free-tier Gemini key after the protocol fixes, using the Archivist's five prompts. The transcript is committed as `05_Noisy_Interrogation_Transcript.md` with its logs.

Item	Value
Provider and models	gemini: 3.8-flash, then 3.5-flash and 3.1-flash-lite as daily quotas ran out
LLM calls	21 (10 source, 10 transmitter, 1 receiver pass)
Transmissions obeying the channel	10 of 10, each 7 words with one noise token
Source entropy	8.12 bits per word
Transmitted entropy	5.85 bits per word
Noise rate and capacity bound	0.143 and 0.857
Lexical retention	0.11
Receiver-judged fidelity	0.91

Item	Value
Human-likeness	0.69, 0 of 10 turns judged HUMAN
FST stress benchmark	4 of 5 Navigator utterances break the FST (80 percent); 5 of 5 in strict mode

Channel

Only about a tenth of the source wording survives the seven-word budget, yet the receiver recovers nine tenths of the meaning. This is the mission's Shannon lesson made concrete: the transmitter packs the high-information words into the available slots and the noise token lands on a filler, so mutual information stays close to the capacity bound even though most symbols are gone. Example transmissions: the three-symbol ritual became "Rise, Hold, sync Fall; desync fails. Quantum", and the rule-only model counterfactual became "Succeeds timestamp formatting; fails nuance sarcasm. Midnight".

Turing

The receiver judged every turn MACHINE while rating the style fairly human. Dense, perfectly packed answers read as machine-like because of their efficiency, which is itself a finding for the Lovelace question.

FST

The canonical session reaches TERMINAL with all three slots filled and two bits of state entropy. Natural phrasing breaks the machine four times out of five, confirming the brittleness the lecture predicts for deterministic transducers.

9. Verification and sign-off

The following checks were run on the final state of the repository.

Check	Result
--selftest (offline, mock provider)	21 of 21 checks pass
--check-signoffs	all four officers signed, cleared for transmission
Live run	completed, transcript and logs committed
Protocol compliance findings P1 to P5	all five applied and covered by self-test checks
Repository visibility	public

10. Limitations and honest notes

- The free-tier key allows five requests per minute and twenty per day per model. The run therefore took over two hours of mostly waiting and spanned three models; the log records which model answered each call.
- Fidelity and human-likeness are ratings by a receiver model, not human raters. Lexical retention and entropy are computed exactly.
- The stress benchmark measures the Navigator's five utterances only. Break rates on other phrasings will differ.
- The Archivist's cross-model comparison was run on ChatGPT only, as the brief states; Gemini and Claude results are listed as future work there.
- The Protocol Specialist's compliance review was written against the simulator before her fixes were applied. The Navigator's benchmark numbers were re-run after the fixes and did not change.

11. Links

- Squad repository: https://github.com/fahadakram009/squadron-bengio-cai
- Mission 2 folder: https://github.com/fahadakram009/squadron-bengio-cai/tree/main/mission-02
- Instructor repository: https://github.com/Unisvet/couse-c-ai
- Course site: https://dlearn-cai.web.app